

ELITE IGCSE MATHEMATICS
EXPERIENCE NOTES

Number Toolkit

Learn the trigger, write the first line, then solve one similar problem while the method is still fresh.

NUMBER TOOLKIT · CH1 · L1

ENG. ESLAM AHMED



The Map

This lesson collapses Number Toolkit into the few exam moves that keep repeating. Read the trigger, copy the first line, then practise the same idea on the next page.

01 Prime factor structure

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

02 Standard form conversion

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

03 Fractions and decimals

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

04 Rounding and bounds

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

05 Convert standard form

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

06 Find the LCM

TRIGGER *"work out", "find", a missing value*

BANK EVIDENCE 106 website questions, 106 checked solutions, 3 local PDFs read.

01 Prime factor structure

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Number Toolkit question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$28 = 2^2 \times 7$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1H Q2

Write both numbers as products

$$28 = 2^2 \times 7$$

Find the LCM

$$\text{LCM} = 2^2 \times 3 \times 5 \times 7 = 420$$

Finish: 420

FORMULA BANK

Write the rule or equation first, then substitute. Do not start with loose calculator work.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

02 Standard form conversion

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Number Toolkit question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$7.8 \times 10^{-4} = 0.00078$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1H Q6

Convert standard form

$$7.8 \times 10^{-4} = 0.00078$$

Convert standard form

$$5.6 \times 10^4 + 7 \times 10^3 = 56000 + 7000 = 63000$$

Finish: (a) 0.00078, (b) 2.25×10^7

FORMULA BANK

Write the rule or equation first, then substitute. Do not start with loose calculator work.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 02

SAME IDEA Use the same first line from Strategy 02.

QUESTION

6 (a) Write 7.8 times 10^{-4} as an ordinary number. (1) (b) Work out $5 \ 6 \ 10 \ 7 \ 10 \ 2 \ 8 \ 10 \ 4 \ 3$
3 . . times + times times - Give your answer in standard form. (2)

YOUR SOLUTION

03 Fractions and decimals

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Number Toolkit question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$3\frac{1}{5} = \frac{16}{5}$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1HR Q2

Simplify fraction

$$3\frac{1}{5} = \frac{16}{5}$$

Simplify fraction

$$\frac{16}{5} \times \frac{21}{8} = \frac{2 \times 21}{5}$$

Finish: $8\frac{2}{5}$

FORMULA BANK

Write the rule or equation first, then substitute. Do not start with loose calculator work.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 03.

QUESTION

2 Show that $315 \text{ times } 258 = 825$

YOUR SOLUTION

04 Rounding and bounds

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Number Toolkit question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$P = \frac{2a - c}{d}$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1H Q16

Estimate the value

$$P = \frac{2a - c}{d}$$

Find upper bound

$$a < 58.45, \quad c \geq 19.5, \quad d \geq 3.55$$

Finish: 27.44

FORMULA BANK

Write the rule or equation first, then substitute. Do not start with loose calculator work.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

05 Convert standard form

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Number Toolkit question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$7.8 \times 10^{-4} = 0.00078$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1H Q6

Convert standard form

$$7.8 \times 10^{-4} = 0.00078$$

Convert standard form

$$5.6 \times 10^4 + 7 \times 10^3 = 56000 + 7000 = 63000$$

Finish: (a) 0.00078, (b) 2.25×10^7

FORMULA BANK

Write the rule or equation first, then substitute. Do not start with loose calculator work.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 05

SAME IDEA Use the same first line from Strategy 05.

QUESTION

6 (a) Write 7.8 times 10^{-4} as an ordinary number. (1) (b) Work out $5 \ 6 \ 10 \ 7 \ 10 \ 2 \ 8 \ 10 \ 4 \ 3$
 $3 \ . \ .$ times + times times - Give your answer in standard form. (2)

YOUR SOLUTION

06 Find the LCM

TRIGGER *"work out", "find", a missing value*

BECOMES A Number Toolkit question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$28 = 2^2 \times 7$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1H Q2

Write both numbers as products

$$28 = 2^2 \times 7$$

Find the LCM

$$\text{LCM} = 2^2 \times 3 \times 5 \times 7 = 420$$

Finish: 420

FORMULA BANK

Write the rule or equation first, then substitute. Do not start with loose calculator work.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Quick Reference

TRIGGER → FIRST MOVE

WHEN YOU SEE	WRITE THIS FIRST
"work out", "show", "hence", a familiar exam phrase	Prime factor structure
"work out", "show", "hence", a familiar exam phrase	Standard form conversion
"work out", "show", "hence", a familiar exam phrase	Fractions and decimals
"work out", "show", "hence", a familiar exam phrase	Rounding and bounds
"work out", "show", "hence", a familiar exam phrase	Convert standard form
"work out", "find", a missing value	Find the LCM
FINAL BOTTOM LINE	Do not try to remember every past-paper wording. Remember the trigger, write the first line, and let the working follow.

ELITE IGCSE MATHEMATICS
EXPERIENCE NOTES

Set Notation & Venn Diagrams

Learn the trigger, write the first line, then solve one similar problem while the method is still fresh.

SET NOTATION & VENN DIAGRAMS · CH1 · L2



ENG. ESLAM AHMED

The Map

This lesson collapses Set Notation & Venn Diagrams into the few exam moves that keep repeating. Read the trigger, copy the first line, then practise the same idea on the next page.

01 Fill the intersection first

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

02 Use union and intersection

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

03 Use complements

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

04 Read probability from the diagram

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

05 Calculate statistic

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BANK EVIDENCE 23 website questions, 23 checked solutions, 3 local PDFs read.

01 Fill the intersection first

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Set Notation & Venn Diagrams question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$B = \{b, l, u, e\}, \quad G = \{g, r, e, y\}, \quad W = \{w, h, i, t, e\}$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1HR Q4

Calculate value

$$B = \{b, l, u, e\}, \quad G = \{g, r, e, y\}, \quad W = \{w, h, i, t, e\}$$

means all the letters that

$$B \cup G = \{b, l, u, e, g, r, y\}$$

Finish: (a)(i) $\{b, l, u, e, g, r, y\}$, (a)(ii) $\{w, h, i, t\}$. Serena is incorrect.

FORMULA BANK

Write the rule or equation first, then substitute. Do not start with loose calculator work.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 01

SAME IDEA Use the same first line from Strategy 01.

QUESTION

4. $B = \{b, l, u, e\}$ $G = \{g, r, e, y\}$ $W = \{w, h, i, t, e\}$ (a) List all the members of the set (i) $B \cup G$ (ii) $W \cap G$ (2) Serena writes down the statement $B \cap G \cap W = \text{empty set}$ (b) Is Serena's statement correct? You must give a reason for your answer. (1)

YOUR SOLUTION

02 Use union and intersection

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Set Notation & Venn Diagrams question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$G \cap S \text{ only} = 11 - 6 = 5$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Specimen 4WM1H Q14

Place the centre value

This is a reasoning step: write one clear sentence, then continue the calculation.

Fill the pair-only regions

$$G \cap S \text{ only} = 11 - 6 = 5$$

Finish: Venn regions: G only 13, S only 9, C only 24, $G \cap S$ only 5, $S \cap C$ only 11, $G \cap C$ only 12, all three 6, none 20. Also 36, 44, 35, and $\frac{18}{53}$.

FORMULA BANK

Write the rule or equation first, then substitute. Do not start with loose calculator work.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 02.

QUESTION

14 100 farmers are asked if they have goats, sheep or chickens. Complete the Venn diagram, find $n(G)$, $n((G \cup S)')$, $n(G' \cap C)$, and the probability a farmer with chickens also has goats.

YOUR SOLUTION

03 Use complements

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Set Notation & Venn Diagrams question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\mathcal{E} = \{20, 21, 22, 23, 24, 25, 26, 27, 28, 29\}$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2021 P1H Q4

Calculate value

$$\mathcal{E} = \{20, 21, 22, 23, 24, 25, 26, 27, 28, 29\}$$

The odd numbers are

$$A = \{21, 23, 25, 27, 29\}$$

Finish: (i) $\{21, 27\}$, (ii) $\{21, 23, 24, 25, 27, 29\}$

FORMULA BANK

Write the rule or equation first, then substitute. Do not start with loose calculator work.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 03.

QUESTION

4 E = {20, 21, 22, 23, 24, 25, 26, 27, 28, 29} A = {odd numbers} B = {multiples of 3} List the members of the set (i) A intersection B (1) (ii) A union B (1)

YOUR SOLUTION

04 Read probability from the diagram

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Set Notation & Venn Diagrams question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$G \cap S \text{ only} = 11 - 6 = 5$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Specimen 4WM1H Q14

Place the centre value

This is a reasoning step: write one clear sentence, then continue the calculation.

Fill the pair-only regions

$$G \cap S \text{ only} = 11 - 6 = 5$$

Finish: Venn regions: G only 13, S only 9, C only 24, $G \cap S$ only 5, $S \cap C$ only 11, $G \cap C$ only 12, all three 6, none 20. Also 36, 44, 35, and $\frac{18}{53}$.

FORMULA BANK

Write the rule or equation first, then substitute. Do not start with loose calculator work.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 04.

QUESTION

14 100 farmers are asked if they have goats, sheep or chickens. Complete the Venn diagram, find $n(G)$, $n((G \cup S)')$, $n(G' \cap C)$, and the probability a farmer with chickens also has goats.

YOUR SOLUTION

05 Calculate statistic

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Set Notation & Venn Diagrams question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$B = \{b, l, u, e\}, \quad G = \{g, r, e, y\}, \quad W = \{w, h, i, t, e\}$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1HR Q4

Calculate value

$$B = \{b, l, u, e\}, \quad G = \{g, r, e, y\}, \quad W = \{w, h, i, t, e\}$$

means all the letters that

$$B \cup G = \{b, l, u, e, g, r, y\}$$

Finish: (a)(i) $\{b, l, u, e, g, r, y\}$, (a)(ii) $\{w, h, i, t\}$. Serena is incorrect.

FORMULA BANK

Write the rule or equation first, then substitute. Do not start with loose calculator work.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 05

SAME IDEA Use the same first line from Strategy 05.

QUESTION

4. $B = \{b, l, u, e\}$ $G = \{g, r, e, y\}$ $W = \{w, h, i, t, e\}$ (a) List all the members of the set (i) $B \cup G$ (ii) $W \cap G$ (2) Serena writes down the statement $B \cap G \cap W = \text{empty set}$ (b) Is Serena's statement correct? You must give a reason for your answer. (1)

YOUR SOLUTION

Quick Reference

TRIGGER → FIRST MOVE

WHEN YOU SEE	WRITE THIS FIRST
"work out", "show", "hence", a familiar exam phrase	Fill the intersection first
"work out", "show", "hence", a familiar exam phrase	Use union and intersection
"work out", "show", "hence", a familiar exam phrase	Use complements
"work out", "show", "hence", a familiar exam phrase	Read probability from the diagram
"work out", "show", "hence", a familiar exam phrase	Calculate statistic

FINAL BOTTOM LINE Do not try to remember every past-paper wording. Remember the

trigger, write the first line, and let the working follow.

ELITE IGCSE MATHEMATICS
EXPERIENCE NOTES

Prime Factors, HCF & LCM

Learn the trigger, write the first line, then solve one similar problem while the method is still fresh.

PRIME FACTORS, HCF & LCM · CH1 · L3



ENG. ESLAM AHMED

The Map

This lesson collapses Prime Factors, HCF & LCM into the few exam moves that keep repeating. Read the trigger, copy the first line, then practise the same idea on the next page.

01 Write prime factor trees

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

02 Find HCF

TRIGGER *"work out", "find", a missing value*

03 Find LCM

TRIGGER *"work out", "find", a missing value*

04 Use HCF/LCM in context

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

05 Find the LCM

TRIGGER *"work out", "find", a missing value*

06 The HCF takes the smaller

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BANK EVIDENCE 25 website questions, 25 checked solutions, 3 local PDFs read.

01 Write prime factor trees

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Prime Factors, HCF & LCM question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$5^{17} \times 5^2 = 5^{19}$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P2HR Q1

Use the index law

$$5^{17} \times 5^2 = 5^{19}$$

Find the LCM

$$800 = 8 \times 100$$

Finish: (a) 5^{19} , (b) $2^5 \times 5^2$

FORMULA BANK

HCF takes common prime powers. LCM takes highest prime powers.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

02 Find HCF

TRIGGER *"work out", "find", a missing value*

BECOMES A Prime Factors, HCF & LCM question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$5^{17} \times 5^2 = 5^{19}$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P2HR Q1

Use the index law

$$5^{17} \times 5^2 = 5^{19}$$

Find the LCM

$$800 = 8 \times 100$$

Finish: (a) 5^{19} , (b) $2^5 \times 5^2$

FORMULA BANK

HCF takes common prime powers. LCM takes highest prime powers.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

03 Find LCM

TRIGGER *"work out", "find", a missing value*

BECOMES A Prime Factors, HCF & LCM question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$A = 2^8 \times 3^5 \times 11^4$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2021 P1HR Q9

Find the LCM

$$A = 2^8 \times 3^5 \times 11^4$$

Find the LCM

$$\text{HCF} = 2^6 \times 3 \times 11^4$$

Finish: (a) $2^6 \times 3 \times 11^4$, (b) $2^9 \times 3^5 \times 11^8$

FORMULA BANK

HCF takes common prime powers. LCM takes highest prime powers.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 03.

QUESTION

9 A = 2 8 times 3 5 times 11 4 B = 2 6 times 3 times 11 8 (a) Find the highest common factor (HCF) of A and B. (2) (b) Find the lowest common multiple (LCM) of 2A and 3B. Give the LCM as a product of powers of its prime factors. (2)

YOUR SOLUTION

04 Use HCF/LCM in context

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Prime Factors, HCF & LCM question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$A = 2^3 \times 3^2 \times 5^2 \times 11$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2021 P2H Q9

Find the LCM

$$A = 2^3 \times 3^2 \times 5^2 \times 11$$

Find the LCM

$$\text{LCM} = 2^4 \times 3^2 \times 5^4 \times 11 \times 13$$

Finish: $2^4 \times 3^2 \times 5^4 \times 11 \times 13$

FORMULA BANK

HCF takes common prime powers. LCM takes highest prime powers.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 04.

QUESTION

9 A = 23 times 32 times 52 times 11 B = 24 times 3 times 54 times 13 Find the lowest common multiple (LCM) of A and B. Give your answer as a product of powers of prime numbers.

YOUR SOLUTION

05 Find the LCM

TRIGGER *"work out", "find", a missing value*

BECOMES A Prime Factors, HCF & LCM question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$28 = 2^2 \times 7$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1H Q2

Write both numbers as products

$$28 = 2^2 \times 7$$

Find the LCM

$$\text{LCM} = 2^2 \times 3 \times 5 \times 7 = 420$$

Finish: 420

FORMULA BANK

HCF takes common prime powers. LCM takes highest prime powers.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

06 The HCF takes the smaller

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Prime Factors, HCF & LCM question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$A = 2^5 \times 5 \times 7^2$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: May 2024 P1HR Q11

Find the LCM

$$A = 2^5 \times 5 \times 7^2$$

Find the LCM

$$5A = 2^5 \times 5^2 \times 7^2$$

Finish: (a) $2^4 \times 5^2 \times 7^2$, (b) $2^{16} \times 5^8 \times 7^{12}$

FORMULA BANK

HCF takes common prime powers. LCM takes highest prime powers.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 06

SAME IDEA Use the same first line from Strategy 06.

QUESTION

11 A B 2 5 7 2 5 7 5 2 3 3 4 (a) Write down the highest common factor (HCF) of 5A and 2B Give your answer as a product of prime factors. (2) A B 2 5 7 2 5 7 5 2 3 3 4 (b) Work out the value of AB 2 Give your answer as a product of prime factors. (2)

YOUR SOLUTION

Quick Reference

TRIGGER → FIRST MOVE

WHEN YOU SEE	WRITE THIS FIRST
"work out", "show", "hence", a familiar exam phrase	Write prime factor trees
"work out", "find", a missing value	Find HCF
"work out", "find", a missing value	Find LCM
"work out", "show", "hence", a familiar exam phrase	Use HCF/LCM in context
"work out", "find", a missing value	Find the LCM
"work out", "show", "hence", a familiar exam phrase	The HCF takes the smaller

FINAL BOTTOM LINE Do not try to remember every past-paper wording. Remember the

trigger, write the first line, and let the working follow.

ELITE IGCSE MATHEMATICS
EXPERIENCE NOTES

Powers, Roots & Standard Form

Learn the trigger, write the first line, then solve one similar problem while the method is still fresh.

POWERS, ROOTS & STANDARD FORM · CH1 · L4



ENG. ESLAM AHMED

The Map

This lesson collapses Powers, Roots & Standard Form into the few exam moves that keep repeating. Read the trigger, copy the first line, then practise the same idea on the next page.

01 Convert standard form

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

02 Use index laws

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

03 Handle negative and fractional powers

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

04 Calculate then return to standard form

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

05 Difference

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

06 Write this in standard form

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BANK EVIDENCE 30 website questions, 30 checked solutions, 3 local PDFs read.

01 Convert standard form

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Powers, Roots & Standard Form question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$7.8 \times 10^{-4} = 0.00078$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1H Q6

Convert standard form

$$7.8 \times 10^{-4} = 0.00078$$

Convert standard form

$$5.6 \times 10^4 + 7 \times 10^3 = 56000 + 7000 = 63000$$

Finish: (a) 0.00078, (b) 2.25×10^7

FORMULA BANK

$a^m a^n = a^{m+n}$, $(a^m)^n = a^{mn}$, standard form is $A \times 10^n$ with $1 \leq A < 10$.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 01.

QUESTION

6 (a) Write 7.8 times 10^{-4} as an ordinary number. (1) (b) Work out $5.6 \times 10^7 \div 10^2 \times 8 \times 10^4 \times 3 \times 10^{-3}$. Give your answer in standard form. (2)

YOUR SOLUTION

02 Use index laws

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Powers, Roots & Standard Form question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

Write the main rule first, then substitute the values.

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Nov 2025 4WM2H Q13

Use index laws

This is a reasoning step: write one clear sentence, then continue the calculation.

Solve inequality

This is a reasoning step: write one clear sentence, then continue the calculation.

Finish: 7.29 times 10^{242}

FORMULA BANK

$a^m a^n = a^{m+n}$, $(a^m)^n = a^{mn}$, standard form is $A \times 10^n$ with $1 \leq A < 10$.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

03 Handle negative and fractional powers

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Powers, Roots & Standard Form question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$7.8 \times 10^{-4} = 0.00078$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1H Q6

Convert standard form

$$7.8 \times 10^{-4} = 0.00078$$

Convert standard form

$$5.6 \times 10^4 + 7 \times 10^3 = 56000 + 7000 = 63000$$

Finish: (a) 0.00078, (b) 2.25×10^7

FORMULA BANK

$a^m a^n = a^{m+n}$, $(a^m)^n = a^{mn}$, standard form is $A \times 10^n$ with $1 \leq A < 10$.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 03.

QUESTION

6 (a) Write 7.8 times 10^{-4} as an ordinary number. (1) (b) Work out 5610710281043
 $3 \dots$ times + times times - Give your answer in standard form. (2)

YOUR SOLUTION

04 Calculate then return to standard form

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Powers, Roots & Standard Form question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$1.357 \times 10^5$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2021 P2HR Q8

The greatest coastline is the

$$1.357 \times 10^5$$

Convert standard form

This is a reasoning step: write one clear sentence, then continue the calculation.

Finish: (a) Pacific Ocean, (b) 9.393×10^4 km

FORMULA BANK

$a^m a^n = a^{m+n}$, $(a^m)^n = a^{mn}$, standard form is $A \times 10^n$ with $1 \leq A < 10$.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 04.

QUESTION

8 The table gives the length of the coastline, in kilometres, of each of five oceans.

Ocean	Length of coastline (km)
Arctic	4.539×10^4
Atlantic	1.119×10^5
Pacific	1.357×10^5
Indian	6.653×10^4
Southern	1.797×10^4

(a) Which ocean has the greatest length of coastline? (1) (b) Calculate the difference between the length of the Atlantic Ocean's coastline and the length of the Southern Ocean's coastline. Give your answer in standard form. km (2)

YOUR SOLUTION

05 Difference

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Powers, Roots & Standard Form question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$1.06 \times 10^8 = 106000000$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P2HR Q10

Atlantic surface area

$$1.06 \times 10^8 = 106000000$$

Indian surface area

$$6.86 \times 10^7 = 68600000$$

Finish: (a) 3.74×10^7 , (b) 11

FORMULA BANK

$a^m a^n = a^{m+n}$, $(a^m)^n = a^{mn}$, standard form is $A \times 10^n$ with $1 \leq A < 10$.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 05.

QUESTION

10 The table shows information about the surface area of each of the world's oceans.

Ocean	Surface area in square kilometres
Pacific	156 times 10 ⁸
Indian	6.86 times 10 ⁷
Southern	2.03 times 10 ⁷
Arctic	1.41 times 10 ⁷
Atlantic	1.06 times 10 ⁸

(a) Work out the difference, in square kilometres, between the surface area of the Atlantic Ocean and the surface area of the Indian Ocean. Give your answer in standard form. square kilometres

(2) The surface area of the Pacific Ocean is k times the surface area of the ...

YOUR SOLUTION

06 Write this in standard form

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Powers, Roots & Standard Form question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$a = 4.2 \times 10^{-24}$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2022 P1H Q8

Convert standard form

$$a = 4.2 \times 10^{-24}$$

Convert standard form

$$ab = (4.2 \times 3) \times 10^{-24+145}$$

Finish: 1.26×10^{122}

FORMULA BANK

$a^m a^n = a^{m+n}$, $(a^m)^n = a^{mn}$, standard form is $A \times 10^n$ with $1 \leq A < 10$.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 06.

QUESTION

8 a = 4.2 times 10^{-24} b = 3 times 10^{145} Work out the value of a times b Give your answer in standard form.

YOUR SOLUTION

Quick Reference

TRIGGER → FIRST MOVE

WHEN YOU SEE	WRITE THIS FIRST
"work out", "show", "hence", a familiar exam phrase	Convert standard form
"work out", "show", "hence", a familiar exam phrase	Use index laws
"work out", "show", "hence", a familiar exam phrase	Handle negative and fractional powers
"work out", "show", "hence", a familiar exam phrase	Calculate then return to standard form
"work out", "show", "hence", a familiar exam phrase	Difference
"work out", "show", "hence", a familiar exam phrase	Write this in standard form

FINAL BOTTOM LINE Do not try to remember every past-paper wording. Remember the trigger, write the first line, and let the working follow.

ELITE IGCSE MATHEMATICS
EXPERIENCE NOTES

Fractions

Learn the trigger, write the first line, then solve one similar problem while the method is still fresh.

FRACTIONS · CH1 · L5

ENG. ESLAM AHMED



The Map

This lesson collapses Fractions into the few exam moves that keep repeating. Read the trigger, copy the first line, then practise the same idea on the next page.

01 Use a common denominator

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

02 Multiply and divide fractions

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

03 Simplify mixed numbers

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

04 Convert the final form

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

05 Simplify fraction

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

06 Convert to improper fractions

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BANK EVIDENCE 14 website questions, 14 checked solutions, 3 local PDFs read.

01 Use a common denominator

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Fractions question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$4\frac{2}{3} = \frac{14}{3}$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P2H Q3

Convert to improper fractions

$$4\frac{2}{3} = \frac{14}{3}$$

Use a common denominator

$$\frac{14}{3} + \frac{19}{5} = \frac{70}{15} + \frac{57}{15}$$

Finish: $8\frac{7}{15}$

FORMULA BANK

Common denominator for addition/subtraction. Multiply straight across. Divide by the reciprocal.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 01.

QUESTION

3 Show that $4^2 3^3 4^5 8^7 15 + =$

YOUR SOLUTION

02 Multiply and divide fractions

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Fractions question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$3\frac{1}{5} = \frac{16}{5}$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1HR Q2

Simplify fraction

$$3\frac{1}{5} = \frac{16}{5}$$

Simplify fraction

$$\frac{16}{5} \times \frac{21}{8} = \frac{2 \times 21}{5}$$

Finish: $8\frac{2}{5}$

FORMULA BANK

Common denominator for addition/subtraction. Multiply straight across. Divide by the reciprocal.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

03 Simplify mixed numbers

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Fractions question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$3\frac{2}{3} = \frac{11}{3}, \quad 2\frac{4}{7} = \frac{18}{7}$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: May 2025 4WM1H Q1

Simplify fraction

$$3\frac{2}{3} = \frac{11}{3}, \quad 2\frac{4}{7} = \frac{18}{7}$$

Simplify fraction

$$\frac{11}{3} \times \frac{18}{7} = \frac{198}{21} = \frac{66}{7}$$

Finish: $9\frac{3}{7}$

FORMULA BANK

Common denominator for addition/subtraction. Multiply straight across. Divide by the reciprocal.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

04 Convert the final form

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Fractions question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$6\frac{3}{4} = \frac{27}{4}$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2022 P2H Q2

Convert to improper fractions

$$6\frac{3}{4} = \frac{27}{4}$$

Divide by multiplying by the

$$\frac{27}{4} \div \frac{18}{7} = \frac{27}{4} \times \frac{7}{18}$$

Finish: $2\frac{5}{8}$

FORMULA BANK

Common denominator for addition/subtraction. Multiply straight across. Divide by the reciprocal.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

05 Simplify fraction

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Fractions question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$3\frac{1}{5} = \frac{16}{5}$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1HR Q2

Simplify fraction

$$3\frac{1}{5} = \frac{16}{5}$$

Simplify fraction

$$\frac{16}{5} \times \frac{21}{8} = \frac{2 \times 21}{5}$$

Finish: $8\frac{2}{5}$

FORMULA BANK

Common denominator for addition/subtraction. Multiply straight across. Divide by the reciprocal.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

06 Convert to improper fractions

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Fractions question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$4\frac{2}{3} = \frac{14}{3}$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P2H Q3

Convert to improper fractions

$$4\frac{2}{3} = \frac{14}{3}$$

Use a common denominator

$$\frac{14}{3} + \frac{19}{5} = \frac{70}{15} + \frac{57}{15}$$

Finish: $8\frac{7}{15}$

FORMULA BANK

Common denominator for addition/subtraction. Multiply straight across. Divide by the reciprocal.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Quick Reference

TRIGGER → FIRST MOVE

WHEN YOU SEE	WRITE THIS FIRST
"work out", "show", "hence", a familiar exam phrase	Use a common denominator
"work out", "show", "hence", a familiar exam phrase	Multiply and divide fractions
"work out", "show", "hence", a familiar exam phrase	Simplify mixed numbers
"work out", "show", "hence", a familiar exam phrase	Convert the final form
"work out", "show", "hence", a familiar exam phrase	Simplify fraction
"work out", "show", "hence", a familiar exam phrase	Convert to improper fractions

FINAL BOTTOM LINE Do not try to remember every past-paper wording. Remember the trigger, write the first line, and let the working follow.

ELITE IGCSE MATHEMATICS
EXPERIENCE NOTES

Percentages

Learn the trigger, write the first line, then solve one similar problem while the method is still fresh.

PERCENTAGES · CH1 · L6

ENG. ESLAM AHMED



The Map

This lesson collapses Percentages into the few exam moves that keep repeating. Read the trigger, copy the first line, then practise the same idea on the next page.

01 Use percentage multiplier

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

02 Reverse a percentage

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

03 Use compound change

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

04 Compare percentage change

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

05 Calculate percentage

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

06 Percentage profit

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BANK EVIDENCE 40 website questions, 40 checked solutions, 3 local PDFs read.

01 Use percentage multiplier

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Percentages question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$100\% - 15\% = 85\%$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1HR Q7

The hat is reduced by

$$100\% - 15\% = 85\%$$

of the normal price

$$0.85 \times \text{normal price} = 20.40$$

Finish: 24 euros.

FORMULA BANK

Percentage multiplier: increase by $r\%$ means $\times (1 + \frac{r}{100})$, decrease means $\times (1 - \frac{r}{100})$.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

02 Reverse a percentage

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Percentages question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$100\% - 15\% = 85\%$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1HR Q7

The hat is reduced by

$$100\% - 15\% = 85\%$$

of the normal price

$$0.85 \times \text{normal price} = 20.40$$

Finish: 24 euros.

FORMULA BANK

Percentage multiplier: increase by $r\%$ means $\times(1 + \frac{r}{100})$, decrease means $\times(1 - \frac{r}{100})$.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

03 Use compound change

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Percentages question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$14 \times 7 \times 5 = 490$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jun 2023 P2HR Q6

Pierre's old weekly pay

This is a reasoning step: write one clear sentence, then continue the calculation.

His new weekly pay is

$$14 \times 7 \times 5 = 490$$

Finish: Pierre's pay increased by 22.5%. Marie's old salary was 24500 euros.

FORMULA BANK

Percentage multiplier: increase by $r\%$ means $\times (1 + \frac{r}{100})$, decrease means $\times (1 - \frac{r}{100})$.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 03

SAME IDEA Use the same first line from Strategy 03.

QUESTION

6 In his previous job, Pierre was paid 400 euros in total for working a 5-day week. In his new job, Pierre is paid 14 euros per hour. In his new job, Pierre works for 7 hours each day for a 5-day week. (a) Work out the percentage increase in the amount that Pierre is paid for a 5-day week. % (4) Marie changes her job. Her salary decreases by 6% Her new salary is 23 030 euros. (b) Work out Marie's salary before she changes her job. euros (3)

YOUR SOLUTION

04 Compare percentage change

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Percentages question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$14 \times 7 \times 5 = 490$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jun 2023 P2HR Q6

Pierre's old weekly pay

This is a reasoning step: write one clear sentence, then continue the calculation.

His new weekly pay is

$$14 \times 7 \times 5 = 490$$

Finish: Pierre's pay increased by 22.5%. Marie's old salary was 24500 euros.

FORMULA BANK

Percentage multiplier: increase by $r\%$ means $\times (1 + \frac{r}{100})$, decrease means $\times (1 - \frac{r}{100})$.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 04

SAME IDEA Use the same first line from Strategy 04.

QUESTION

6 In his previous job, Pierre was paid 400 euros in total for working a 5-day week. In his new job, Pierre is paid 14 euros per hour. In his new job, Pierre works for 7 hours each day for a 5-day week. (a) Work out the percentage increase in the amount that Pierre is paid for a 5-day week. % (4) Marie changes her job. Her salary decreases by 6% Her new salary is 23 030 euros. (b) Work out Marie's salary before she changes her job. euros (3)

YOUR SOLUTION

05 Calculate percentage

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Percentages question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$545 - 500 = 45$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P2H Q8

Convert standard form

$$545 - 500 = 45$$

Convert standard form

$$592 - 545 = 47$$

Finish: Brigitte is not correct; the discount is 168 euros.

FORMULA BANK

Percentage multiplier: increase by $r\%$ means $\times(1 + \frac{r}{100})$, decrease means $\times(1 - \frac{r}{100})$.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 05

SAME IDEA Use the same first line from Strategy 05.

QUESTION

8 The table shows the cost, in euros, of Brigitte's car insurance in each of the years 2016, 2017 and 2018

Year	2016	2017	2018
Cost of insurance (euros)	500	545	592

Brigitte says, "The percentage increase in the cost of my car insurance from 2017 to 2018 is more than the percentage increase in the cost of my car insurance from 2016 to 2017"

(a) Is Brigitte correct? You must show how you get your answer. (4)

Henri wants to insure his car. He gets a discount of 15% off the normal price. Henri pays 952 euros for his ...

YOUR SOLUTION

06 Percentage profit

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Percentages question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$520 - 465 = 55$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2021 P2H Q3

Profit

$$520 - 465 = 55$$

Percentage profit

$$\frac{55}{465} \times 100 = 11.827 \dots$$

Finish: (a) 11.8%, (b) \$484

FORMULA BANK

Percentage multiplier: increase by $r\%$ means $\times (1 + \frac{r}{100})$, decrease means $\times (1 - \frac{r}{100})$.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 06

SAME IDEA Use the same first line from Strategy 06.

QUESTION

3 Gladys buys a table for \$465 to sell in her shop. She sells the table for \$520 (a) Work out the percentage profit that Gladys makes from the sale of the table. Give your answer correct to 3 significant figures. % (3) Gladys has a sale in her shop. She decreases all the normal prices by 12% The normal price of an armchair was \$550 (b) Work out the sale price of the armchair. \$ (3)

YOUR SOLUTION

Quick Reference

TRIGGER → FIRST MOVE

WHEN YOU SEE	WRITE THIS FIRST
"work out", "show", "hence", a familiar exam phrase	Use percentage multiplier
"work out", "show", "hence", a familiar exam phrase	Reverse a percentage
"work out", "show", "hence", a familiar exam phrase	Use compound change
"work out", "show", "hence", a familiar exam phrase	Compare percentage change
"work out", "show", "hence", a familiar exam phrase	Calculate percentage
"work out", "show", "hence", a familiar exam phrase	Percentage profit

FINAL BOTTOM LINE Do not try to remember every past-paper wording. Remember the trigger, write the first line, and let the working follow.

ELITE IGCSE MATHEMATICS
EXPERIENCE NOTES

Compound Interest & Depreciation

Learn the trigger, write the first line, then solve one similar problem while the method is still fresh.

COMPOUND INTEREST & DEPRECIATION · CH1 · L7

ENG. ESLAM AHMED



The Map

This lesson collapses Compound Interest & Depreciation into the few exam moves that keep repeating. Read the trigger, copy the first line, then practise the same idea on the next page.

01 Use percentage multiplier

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

02 Reverse a percentage

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

03 Use compound change

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

04 Compare percentage change

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

05 Use compound interest

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

06 The multiplier for compound interest

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BANK EVIDENCE 32 website questions, 32 checked solutions, 3 local PDFs read.

01 Use percentage multiplier

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Compound Interest & Depreciation question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

| $0.85V$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2021 P2HR Q14

Let the original house value

This is a reasoning step: write one clear sentence, then continue the calculation.

Use compound interest

| $0.85V$

Finish: 10%

FORMULA BANK

Repeated percentage change: $A = P(1 \pm r/100)^n$.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 01.

QUESTION

14 Simon bought a house at the beginning of 2018. The value of Simon's house had decreased by 15% by the end of 2018. The house increased in value during both 2019 and 2020. The percentage increases in the value of the house during 2019 and 2020 were the same. The value of Simon's house at the end of 2020 was 2.85% greater than the amount he paid for his house at the beginning of 2018. Calculate the percentage increase in the value of the house during 2019 %

YOUR SOLUTION

02 Reverse a percentage

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Compound Interest & Depreciation question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

| $0.85V$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2021 P2HR Q14

Let the original house value

This is a reasoning step: write one clear sentence, then continue the calculation.

Use compound interest

| $0.85V$

Finish: 10%

FORMULA BANK

Repeated percentage change: $A = P(1 \pm r/100)^n$.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

03 Use compound change

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Compound Interest & Depreciation question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$6000(1.015)^2 = 6181.35$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1H Q11

Use compound interest

$$6000(1.015)^2 = 6181.35$$

Let the third year interest

$$6181.35 \left(1 + \frac{r}{100}\right) = 6311.16$$

Finish: 2.1%

FORMULA BANK

Repeated percentage change: $A = P(1 \pm r/100)^n$.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 03

SAME IDEA Use the same first line from Strategy 03.

QUESTION

11 Max invests \$6000 in a savings account for 3 years. The account pays compound interest at a rate of 1.5% per year for the first 2 years. The compound interest rate changes for the third year. At the end of 3 years, there is a total of \$6311.16 in the account. Work out the compound interest rate for the third year. Give your answer correct to 1 decimal place. %

YOUR SOLUTION

04 Compare percentage change

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Compound Interest & Depreciation question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$2020 \text{ value} = \frac{634400}{1.04} = 610000$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jun 2022 P1H Q6

Use compound interest

$$2020 \text{ value} = \frac{634400}{1.04} = 610000$$

Use compound interest

This is a reasoning step: write one clear sentence, then continue the calculation.

Finish: (a) No, it stayed in band B. (b) 27.75%

FORMULA BANK

Repeated percentage change: $A = P(1 \pm r/100)^n$.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 04.

6 Asha bought an apartment. The table gives information about the value of apartments, in euros, and the annual service charge band. Value (x euros) Service charge band
 $x \geq 700\,000$ A $600\,000 < x < 700\,000$ B $500\,000 < x < 600\,000$ C $400\,000 < x < 500\,000$ D $0 < x < 400\,000$ E
 In 2021, the value of Asha's apartment was 634 400 euros. The value of Asha's apartment had increased by 4% from its value in 2020 (a) Has the annual service charge band changed for Asha's apartment? Show your working clearly.
 (3) Pam bought a ...

05 Use compound interest

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Compound Interest & Depreciation question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$6000(1.015)^2 = 6181.35$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1H Q11

Use compound interest

$$6000(1.015)^2 = 6181.35$$

Let the third year interest

$$6181.35 \left(1 + \frac{r}{100}\right) = 6311.16$$

Finish: 2.1%

FORMULA BANK

Repeated percentage change: $A = P(1 \pm r/100)^n$.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 05

SAME IDEA Use the same first line from Strategy 05.

QUESTION

11 Max invests \$6000 in a savings account for 3 years. The account pays compound interest at a rate of 1.5% per year for the first 2 years. The compound interest rate changes for the third year. At the end of 3 years, there is a total of \$6311.16 in the account. Work out the compound interest rate for the third year. Give your answer correct to 1 decimal place. %

YOUR SOLUTION

06 The multiplier for compound interest

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Compound Interest & Depreciation question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

	1.025
--	-------

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2022 P2HR Q5

The multiplier for compound interest

1.025

Use compound interest

$$7200(1.025)^3 = 7753.6125$$

Finish: 7754 dollars

FORMULA BANK

Repeated percentage change: $A = P(1 \pm r/100)^n$.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 06.

QUESTION

5 Shane invests 7200 dollars for 3 years in a savings account. He gets 2.5% per year compound interest. How much money will Shane have in his savings account at the end of 3 years? Give your answer to the nearest dollar. dollars

YOUR SOLUTION

Quick Reference

TRIGGER → FIRST MOVE

WHEN YOU SEE	WRITE THIS FIRST
"work out", "show", "hence", a familiar exam phrase	Use percentage multiplier
"work out", "show", "hence", a familiar exam phrase	Reverse a percentage
"work out", "show", "hence", a familiar exam phrase	Use compound change
"work out", "show", "hence", a familiar exam phrase	Compare percentage change
"work out", "show", "hence", a familiar exam phrase	Use compound interest
"work out", "show", "hence", a familiar exam phrase	The multiplier for compound interest
FINAL BOTTOM LINE	Do not try to remember every past-paper wording. Remember the trigger, write the first line, and let the working follow.

ELITE IGCSE MATHEMATICS
EXPERIENCE NOTES

Fractions, Decimals & Percentages

Learn the trigger, write the first line, then solve one similar problem while the method is still fresh.

FRACTIONS, DECIMALS & PERCENTAGES · CH1 · L8



ENG. ESLAM AHMED

The Map

This lesson collapses Fractions, Decimals & Percentages into the few exam moves that keep repeating. Read the trigger, copy the first line, then practise the same idea on the next page.

01 Convert fraction to decimal

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

02 Convert decimal to percentage

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

03 Use percentage multiplier

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

04 Compare all values in one form

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

05 Simplify fraction

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

06 Subtract

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BANK EVIDENCE 1 website questions, 1 checked solutions, 3 local PDFs read.

01 Convert fraction to decimal

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Fractions, Decimals & Percentages question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$y = 0.4\dot{x}$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Nov 2021 P1H Q18

Let

$$y = 0.4\dot{x}$$

Simplify fraction

$$y = 0.4xxx \dots$$

Finish: $\frac{x+36}{90}$

FORMULA BANK

Write the rule or equation first, then substitute. Do not start with loose calculator work.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 01.

QUESTION

18 $0.4x$ is a recurring decimal. x is a whole number such that $1 < x < 9$. Find, in terms of x , the recurring decimal $0.4x$ as a fraction. Give your fraction in its simplest form. Show clear algebraic working.

YOUR SOLUTION

02 Convert decimal to percentage

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Fractions, Decimals & Percentages question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$y = 0.4x$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Nov 2021 P1H Q18

Let

$$y = 0.4x$$

Simplify fraction

$$y = 0.4xxx \dots$$

Finish: $\frac{x+36}{90}$

FORMULA BANK

Write the rule or equation first, then substitute. Do not start with loose calculator work.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

03 Use percentage multiplier

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Fractions, Decimals & Percentages question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$y = 0.4x$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Nov 2021 P1H Q18

Let

$$y = 0.4x$$

Simplify fraction

$$y = 0.4xxx \dots$$

Finish: $\frac{x+36}{90}$

FORMULA BANK

Write the rule or equation first, then substitute. Do not start with loose calculator work.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 03.

QUESTION

18 $0.4x$ is a recurring decimal. x is a whole number such that $1 < x < 9$. Find, in terms of x , the recurring decimal $0.4x$ as a fraction. Give your fraction in its simplest form. Show clear algebraic working.

YOUR SOLUTION

04 Compare all values in one form

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Fractions, Decimals & Percentages question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$y = 0.4x$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Nov 2021 P1H Q18

Let

$$y = 0.4x$$

Simplify fraction

$$y = 0.4xxx \dots$$

Finish: $\frac{x+36}{90}$

FORMULA BANK

Write the rule or equation first, then substitute. Do not start with loose calculator work.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

05 Simplify fraction

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Fractions, Decimals & Percentages question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$y = 0.4x$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Nov 2021 P1H Q18

Let

$$y = 0.4x$$

Simplify fraction

$$y = 0.4xxx \dots$$

Finish: $\frac{x+36}{90}$

FORMULA BANK

Write the rule or equation first, then substitute. Do not start with loose calculator work.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

06 Subtract

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Fractions, Decimals & Percentages question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$y = 0.4x$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Nov 2021 P1H Q18

Let

$$y = 0.4x$$

Simplify fraction

$$y = 0.4xxx \dots$$

Finish: $\frac{x+36}{90}$

FORMULA BANK

Write the rule or equation first, then substitute. Do not start with loose calculator work.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Quick Reference

TRIGGER → FIRST MOVE

WHEN YOU SEE	WRITE THIS FIRST
"work out", "show", "hence", a familiar exam phrase	Convert fraction to decimal
"work out", "show", "hence", a familiar exam phrase	Convert decimal to percentage
"work out", "show", "hence", a familiar exam phrase	Use percentage multiplier
"work out", "show", "hence", a familiar exam phrase	Compare all values in one form
"work out", "show", "hence", a familiar exam phrase	Simplify fraction
"work out", "show", "hence", a familiar exam phrase	Subtract

FINAL BOTTOM LINE Do not try to remember every past-paper wording. Remember the trigger, write the first line, and let the working follow.

ELITE IGCSE MATHEMATICS
EXPERIENCE NOTES

Rounding, Estimation & Bounds

Learn the trigger, write the first line, then solve one similar problem while the method is still fresh.

ROUNDING, ESTIMATION & BOUNDS · CH1 · L9



ENG. ESLAM AHMED

The Map

This lesson collapses Rounding, Estimation & Bounds into the few exam moves that keep repeating. Read the trigger, copy the first line, then practise the same idea on the next page.

01 Round to the requested accuracy

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

02 Find lower and upper bounds

TRIGGER *"work out", "find", a missing value*

03 Choose bounds for largest value

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

04 Choose bounds for smallest value

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

05 Estimate the value

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

06 Find upper bound

TRIGGER *"work out", "find", a missing value*

BANK EVIDENCE 37 website questions, 37 checked solutions, 3 local PDFs read.

01 Round to the requested accuracy

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Rounding, Estimation & Bounds question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$X = \text{area of rectangle} - \text{area of square}$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jun 2025 P2H Q22

The shaded area is

$$X = \text{area of rectangle} - \text{area of square}$$

Bounds

$$11.25 \leq AB < 11.75$$

Finish: $X = 90 \text{ cm}^2$ to a suitable degree of accuracy

FORMULA BANK

Half the rounding unit gives lower/upper bounds. Choose upper/lower to make the final expression largest or smallest.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 01

SAME IDEA Use the same first line from Strategy 01.

QUESTION

22 The diagram shows a square inside rectangle ABCD. A B D C Diagram NOT accurately drawn. The total area of the region shown shaded in the diagram is $X \text{ cm}^2$. $AB = 11.5 \text{ cm}$ correct to the nearest 0.5 cm . $BC = 9.2 \text{ cm}$ correct to 2 significant figures. Side of square = 4.1 cm correct to 2 significant figures. By considering bounds, work out the value of X to a suitable degree of accuracy. Show your working clearly. $X =$

YOUR SOLUTION

02 Find lower and upper bounds

TRIGGER *"work out", "find", a missing value*

BECOMES A Rounding, Estimation & Bounds question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$4.25 \leq w < 4.35$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1HR Q11

Estimate the value

$$4.25 \leq w < 4.35$$

Find upper bound

This is a reasoning step: write one clear sentence, then continue the calculation.

Finish: (a) 4.35 kg, (b) 4.25 kg, (c) 8.15

FORMULA BANK

Half the rounding unit gives lower/upper bounds. Choose upper/lower to make the final expression largest or smallest.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 02

SAME IDEA Use the same first line from Strategy 02.

QUESTION

11 The weight of a cat is 4.3 kg correct to 2 significant figures. (a) Write down the upper bound of the weight of the cat. kg (1) (b) Write down the lower bound of the weight of the cat. kg (1) G = e - f e = 17 correct to the nearest integer f = 9.4 correct to one decimal place (c) Work out the upper bound for the value of G. (2)

YOUR SOLUTION

03 Choose bounds for largest value

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Rounding, Estimation & Bounds question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$P = \frac{2a - c}{d}$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1H Q16

Estimate the value

$$P = \frac{2a - c}{d}$$

Find upper bound

$$a < 58.45, \quad c \geq 19.5, \quad d \geq 3.55$$

Finish: 27.44

FORMULA BANK

Half the rounding unit gives lower/upper bounds. Choose upper/lower to make the final expression largest or smallest.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

04 Choose bounds for smallest value

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Rounding, Estimation & Bounds question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$P = \frac{2a - c}{d}$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1H Q16

Estimate the value

$$P = \frac{2a - c}{d}$$

Find upper bound

$$a < 58.45, \quad c \geq 19.5, \quad d \geq 3.55$$

Finish: 27.44

FORMULA BANK

Half the rounding unit gives lower/upper bounds. Choose upper/lower to make the final expression largest or smallest.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

05 Estimate the value

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Rounding, Estimation & Bounds question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$P = \frac{2a - c}{d}$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1H Q16

Estimate the value

$$P = \frac{2a - c}{d}$$

Find upper bound

$$a < 58.45, \quad c \geq 19.5, \quad d \geq 3.55$$

Finish: 27.44

FORMULA BANK

Half the rounding unit gives lower/upper bounds. Choose upper/lower to make the final expression largest or smallest.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

06 Find upper bound

TRIGGER *"work out", "find", a missing value*

BECOMES A Rounding, Estimation & Bounds question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$P = \frac{2a - c}{d}$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1H Q16

Estimate the value

$$P = \frac{2a - c}{d}$$

Find upper bound

$$a < 58.45, \quad c \geq 19.5, \quad d \geq 3.55$$

Finish: 27.44

FORMULA BANK

Half the rounding unit gives lower/upper bounds. Choose upper/lower to make the final expression largest or smallest.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 06

SAME IDEA Use the same first line from Strategy 06.

QUESTION

16 $P = 2a \times c \times d - a = 58.4$ correct to 3 significant figures. $c = 20$ correct to 2 significant figures. $d = 3.6$ correct to 2 significant figures. Work out the upper bound for the value of P . Show your working clearly. Give your answer correct to 2 decimal places.

YOUR SOLUTION

Quick Reference

TRIGGER → FIRST MOVE

WHEN YOU SEE	WRITE THIS FIRST
"work out", "show", "hence", a familiar exam phrase	Round to the requested accuracy
"work out", "find", a missing value	Find lower and upper bounds
"work out", "show", "hence", a familiar exam phrase	Choose bounds for largest value
"work out", "show", "hence", a familiar exam phrase	Choose bounds for smallest value
"work out", "show", "hence", a familiar exam phrase	Estimate the value
"work out", "find", a missing value	Find upper bound
FINAL BOTTOM LINE	Do not try to remember every past-paper wording. Remember the

trigger, write the first line, and let the working follow.

ELITE IGCSE MATHEMATICS
EXPERIENCE NOTES

Surds

Learn the trigger, write the first line, then solve one similar problem while the method is still fresh.

SURDS · CH1 · L10

ENG. ESLAM AHMED



The Map

This lesson collapses Surds into the few exam moves that keep repeating. Read the trigger, copy the first line, then practise the same idea on the next page.

01 Simplify by square factors

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

02 Add and subtract like surds

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

03 Expand surd brackets

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

04 Rationalise the denominator

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

05 Simplify surd

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

06 Use trigonometry

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BANK EVIDENCE 14 website questions, 14 checked solutions, 3 local PDFs read.

01 Simplify by square factors

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Surds question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$x^2(2\sqrt{5} - 3) = 13 + 6\sqrt{5}$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jun 2023 P2H Q23

The volume is

$$x^2(2\sqrt{5} - 3) = 13 + 6\sqrt{5}$$

Simplify surd

$$x^2 = \frac{13 + 6\sqrt{5}}{2\sqrt{5} - 3}$$

Finish: $2 + \sqrt{5}$

FORMULA BANK

Simplify by square factors. Rationalise using the conjugate when the denominator has a surd.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 01

SAME IDEA Use the same first line from Strategy 01.

QUESTION

23 The diagram shows a cuboid with a square cross section. x cm x cm $(25 - 3)$ cm
Diagram NOT accurately drawn The volume of the cuboid is $(13 + 65)$ cm³ Without using a calculator, find the value of x Give your answer in the form $a + b$ where a and b are integers. Show your working clearly. $x =$

YOUR SOLUTION

02 Add and subtract like surds

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Surds question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\text{width} = \frac{18}{\sqrt{7} + 1}$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P2H Q20

Simplify surd

$$\text{width} = \frac{18}{\sqrt{7} + 1}$$

Rationalise the denominator

$$\frac{18}{\sqrt{7} + 1} \times \frac{\sqrt{7} - 1}{\sqrt{7} - 1} = \frac{18(\sqrt{7} - 1)}{7 - 1}$$

Finish: $3\sqrt{7} - 3$ cm

FORMULA BANK

Simplify by square factors. Rationalise using the conjugate when the denominator has a surd.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

03 Expand surd brackets

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Surds question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$(8 - \sqrt{x})(5 + \sqrt{x}) = 40 + 8\sqrt{x} - 5\sqrt{x} - x$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2021 P1H Q18

Simplify surd

$$(8 - \sqrt{x})(5 + \sqrt{x}) = 40 + 8\sqrt{x} - 5\sqrt{x} - x$$

Simplify surd

$$y\sqrt{x} + 21$$

Finish: $x = 19, y = 3$

FORMULA BANK

Simplify by square factors. Rationalise using the conjugate when the denominator has a surd.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 03.

QUESTION

18 Given that $(8 - x)(5 + x) = yx + 21$ where x is a prime number and y is an integer, find the value of x and the value of y . Show each stage of your working clearly. $x = y =$

YOUR SOLUTION

04 Rationalise the denominator

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Surds question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\text{width} = \frac{18}{\sqrt{7} + 1}$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P2H Q20

Simplify surd

$$\text{width} = \frac{18}{\sqrt{7} + 1}$$

Rationalise the denominator

$$\frac{18}{\sqrt{7} + 1} \times \frac{\sqrt{7} - 1}{\sqrt{7} - 1} = \frac{18(\sqrt{7} - 1)}{7 - 1}$$

Finish: $3\sqrt{7} - 3$ cm

FORMULA BANK

Simplify by square factors. Rationalise using the conjugate when the denominator has a surd.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 04.

QUESTION

20 The area of a rectangle is 18 cm^2 . The length of the rectangle is $7\frac{1}{2} \text{ cm}$. Without using a calculator and showing each stage of your working, find the width of the rectangle. Give your answer in the form $a + b + c$ where a , b and c are integers. cm

YOUR SOLUTION

05 Simplify surd

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Surds question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\text{width} = \frac{18}{\sqrt{7} + 1}$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P2H Q20

Simplify surd

$$\text{width} = \frac{18}{\sqrt{7} + 1}$$

Rationalise the denominator

$$\frac{18}{\sqrt{7} + 1} \times \frac{\sqrt{7} - 1}{\sqrt{7} - 1} = \frac{18(\sqrt{7} - 1)}{7 - 1}$$

Finish: $3\sqrt{7} - 3$ cm

FORMULA BANK

Simplify by square factors. Rationalise using the conjugate when the denominator has a surd.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 05

SAME IDEA Use the same first line from Strategy 05.

QUESTION

20 The area of a rectangle is 18 cm^2 . The length of the rectangle is $7\frac{1}{2} \text{ cm}$. Without using a calculator and showing each stage of your working, find the width of the rectangle. Give your answer in the form $a + b + c$ where a , b and c are integers. cm

YOUR SOLUTION

06 Use trigonometry

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Surds question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$x^2(2\sqrt{5} - 3)$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jun 2023 P2H Q23

The cuboid has square cross

$$x^2(2\sqrt{5} - 3)$$

Given

$$x^2(2\sqrt{5} - 3) = 13 + 6\sqrt{5}$$

Finish: $2 + \sqrt{5}$

FORMULA BANK

Simplify by square factors. Rationalise using the conjugate when the denominator has a surd.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 06

SAME IDEA Use the same first line from Strategy 06.

QUESTION

23 The diagram shows a cuboid with a square cross section. x cm x cm $(25 - 3)$ cm
Diagram NOT accurately drawn The volume of the cuboid is $(13 + 65)$ cm³ Without using a calculator, find the value of x Give your answer in the form $a + b$ where a and b are integers. Show your working clearly. $x =$

YOUR SOLUTION

Quick Reference

TRIGGER → FIRST MOVE

WHEN YOU SEE	WRITE THIS FIRST
"work out", "show", "hence", a familiar exam phrase	Simplify by square factors
"work out", "show", "hence", a familiar exam phrase	Add and subtract like surds
"work out", "show", "hence", a familiar exam phrase	Expand surd brackets
"work out", "show", "hence", a familiar exam phrase	Rationalise the denominator
"work out", "show", "hence", a familiar exam phrase	Simplify surd
"work out", "show", "hence", a familiar exam phrase	Use trigonometry

FINAL BOTTOM LINE Do not try to remember every past-paper wording. Remember the trigger, write the first line, and let the working follow.

ELITE IGCSE MATHEMATICS
EXPERIENCE NOTES

Using a Calculator

Learn the trigger, write the first line, then solve one similar problem while the method is still fresh.

USING A CALCULATOR · CH1 · L11

ENG. ESLAM AHMED



The Map

This lesson collapses Using a Calculator into the few exam moves that keep repeating. Read the trigger, copy the first line, then practise the same idea on the next page.

01 Type the whole expression safely

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

02 Use brackets for fractions and powers

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

03 Read standard form display

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

04 Round only at the end

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

05 Convert standard form

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

06 Estimate the value

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BANK EVIDENCE 67 website questions, 67 checked solutions, 3 local PDFs read.

01 Type the whole expression safely

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Using a Calculator question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$1.06 \times 10^8 = 106000000$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P2HR Q10

Atlantic surface area

$$1.06 \times 10^8 = 106000000$$

Indian surface area

$$6.86 \times 10^7 = 68600000$$

Finish: (a) 3.74×10^7 , (b) 11

FORMULA BANK

Write the rule or equation first, then substitute. Do not start with loose calculator work.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 01

SAME IDEA Use the same first line from Strategy 01.

QUESTION

10 The table shows information about the surface area of each of the world's oceans.

Ocean	Surface area in square kilometres
Pacific	1.56×10^8
Indian	6.86×10^7
Southern	2.03×10^7
Arctic	1.41×10^7
Atlantic	1.06×10^8

(a) Work out the difference, in square kilometres, between the surface area of the Atlantic Ocean and the surface area of the Indian Ocean. Give your answer in standard form. square kilometres

(2) The surface area of the Pacific Ocean is k times the surface area of the ...

YOUR SOLUTION

02 Use brackets for fractions and powers

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Using a Calculator question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$7.8 \times 10^{-4} = 0.00078$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1H Q6

Convert standard form

$$7.8 \times 10^{-4} = 0.00078$$

Convert standard form

$$5.6 \times 10^4 + 7 \times 10^3 = 56000 + 7000 = 63000$$

Finish: (a) 0.00078, (b) 2.25×10^7

FORMULA BANK

Write the rule or equation first, then substitute. Do not start with loose calculator work.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 02

SAME IDEA Use the same first line from Strategy 02.

QUESTION

6 (a) Write 7.8 times 10^{-4} as an ordinary number. (1) (b) Work out $5 \frac{6}{10} \times 7 \frac{10}{10} \times 2 \frac{8}{10} \times 4 \frac{3}{10}$. . times + times times - Give your answer in standard form. (2)

YOUR SOLUTION

03 Read standard form display

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Using a Calculator question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$7.8 \times 10^{-4} = 0.00078$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1H Q6

Convert standard form

$$7.8 \times 10^{-4} = 0.00078$$

Convert standard form

$$5.6 \times 10^4 + 7 \times 10^3 = 56000 + 7000 = 63000$$

Finish: (a) 0.00078, (b) 2.25×10^7

FORMULA BANK

Write the rule or equation first, then substitute. Do not start with loose calculator work.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 03

SAME IDEA Use the same first line from Strategy 03.

QUESTION

6 (a) Write 7.8 times 10^{-4} as an ordinary number. (1) (b) Work out $5 \frac{6}{10} \times 7 \frac{10}{10} \times 2 \frac{8}{10} \times 4 \frac{3}{10}$. . times + times times - Give your answer in standard form. (2)

YOUR SOLUTION

04 Round only at the end

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Using a Calculator question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$P = \frac{2a - c}{d}$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1H Q16

Estimate the value

$$P = \frac{2a - c}{d}$$

Find upper bound

$$a < 58.45, \quad c \geq 19.5, \quad d \geq 3.55$$

Finish: 27.44

FORMULA BANK

Write the rule or equation first, then substitute. Do not start with loose calculator work.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 04

SAME IDEA Use the same first line from Strategy 04.

QUESTION

16 $P = 2a \times c \times d - a = 58.4$ correct to 3 significant figures. $c = 20$ correct to 2 significant figures. $d = 3.6$ correct to 2 significant figures. Work out the upper bound for the value of P . Show your working clearly. Give your answer correct to 2 decimal places.

YOUR SOLUTION

05 Convert standard form

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Using a Calculator question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$7.8 \times 10^{-4} = 0.00078$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1H Q6

Convert standard form

$$7.8 \times 10^{-4} = 0.00078$$

Convert standard form

$$5.6 \times 10^4 + 7 \times 10^3 = 56000 + 7000 = 63000$$

Finish: (a) 0.00078, (b) 2.25×10^7

FORMULA BANK

Write the rule or equation first, then substitute. Do not start with loose calculator work.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 05.

QUESTION

6 (a) Write 7.8 times 10^{-4} as an ordinary number. (1) (b) Work out $5 \ 6 \ 10 \ 7 \ 10 \ 2 \ 8 \ 10 \ 4 \ 3$
3 . . times + times times - Give your answer in standard form. (2)

YOUR SOLUTION

06 Estimate the value

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Using a Calculator question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$P = \frac{2a - c}{d}$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1H Q16

Estimate the value

$$P = \frac{2a - c}{d}$$

Find upper bound

$$a < 58.45, \quad c \geq 19.5, \quad d \geq 3.55$$

Finish: 27.44

FORMULA BANK

Write the rule or equation first, then substitute. Do not start with loose calculator work.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 06

SAME IDEA Use the same first line from Strategy 06.

QUESTION

16 $P = 2a c d - a = 58.4$ correct to 3 significant figures. $c = 20$ correct to 2 significant figures. $d = 3.6$ correct to 2 significant figures. Work out the upper bound for the value of P . Show your working clearly. Give your answer correct to 2 decimal places.

YOUR SOLUTION

Quick Reference

TRIGGER → FIRST MOVE

WHEN YOU SEE	WRITE THIS FIRST
"work out", "show", "hence", a familiar exam phrase	Type the whole expression safely
"work out", "show", "hence", a familiar exam phrase	Use brackets for fractions and powers
"work out", "show", "hence", a familiar exam phrase	Read standard form display
"work out", "show", "hence", a familiar exam phrase	Round only at the end
"work out", "show", "hence", a familiar exam phrase	Convert standard form
"work out", "show", "hence", a familiar exam phrase	Estimate the value
FINAL BOTTOM LINE	Do not try to remember every past-paper wording. Remember the trigger, write the first line, and let the working follow.

ELITE IGCSE MATHEMATICS
EXPERIENCE NOTES

Ratio Toolkit

Learn the trigger, write the first line, then solve one similar problem while the method is still fresh.

RATIO TOOLKIT · CH1 · L12

ENG. ESLAM AHMED



The Map

This lesson collapses Ratio Toolkit into the few exam moves that keep repeating. Read the trigger, copy the first line, then practise the same idea on the next page.

01 Find one share

TRIGGER *"work out", "find", a missing value*

02 Use total parts

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

03 Convert mixed information to parts

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

04 Check the order of the ratio

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

05 Split the ratio

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

06 Total sailing

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BANK EVIDENCE 5 website questions, 5 checked solutions, 3 local PDFs read.

01 Find one share

TRIGGER *"work out", "find", a missing value*

BECOMES A Ratio Toolkit question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\text{girls} = 120 \times \frac{3}{8} = 45$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1H Q5

Split the ratio

$$\text{girls} = 120 \times \frac{3}{8} = 45$$

Split the ratio

$$\frac{16}{25} \times 75 = 48$$

Finish: 57

FORMULA BANK

Add ratio parts, find one part, then multiply. Keep the order of the ratio fixed.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 01

SAME IDEA Use the same first line from Strategy 01.

QUESTION

5 120 children go on an activity holiday. The ratio of the number of girls to the number of boys is 3 : 5 On Sunday, all the children either go sailing or go climbing. 16 25 of the boys go climbing. Twice as many girls go sailing as go climbing. Work out how many children go sailing on Sunday.

YOUR SOLUTION

02 Use total parts

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Ratio Toolkit question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\text{girls} = 120 \times \frac{3}{8} = 45$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1H Q5

Split the ratio

$$\text{girls} = 120 \times \frac{3}{8} = 45$$

Split the ratio

$$\frac{16}{25} \times 75 = 48$$

Finish: 57

FORMULA BANK

Add ratio parts, find one part, then multiply. Keep the order of the ratio fixed.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

03 Convert mixed information to parts

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Ratio Toolkit question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$9 \text{ parts} = 36$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: May 2021 P1H Q4

Ducks pigs

This is a reasoning step: write one clear sentence, then continue the calculation.

Split the ratio

$$9 \text{ parts} = 36$$

Finish: 70

FORMULA BANK

Add ratio parts, find one part, then multiply. Keep the order of the ratio fixed.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 03.

QUESTION

4 On a farm there are chickens, ducks and pigs. The ratio of the number of chickens to the number of ducks is 7 : 2 The ratio of the number of ducks to the number of pigs is 5 : 9 There are 36 pigs on the farm. Work out the number of chickens on the farm.

YOUR SOLUTION

04 Check the order of the ratio

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Ratio Toolkit question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\text{girls} = 120 \times \frac{3}{8} = 45$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1H Q5

Split the ratio

$$\text{girls} = 120 \times \frac{3}{8} = 45$$

Split the ratio

$$\frac{16}{25} \times 75 = 48$$

Finish: 57

FORMULA BANK

Add ratio parts, find one part, then multiply. Keep the order of the ratio fixed.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 04

SAME IDEA Use the same first line from Strategy 04.

QUESTION

5 120 children go on an activity holiday. The ratio of the number of girls to the number of boys is 3 : 5 On Sunday, all the children either go sailing or go climbing. 16 25 of the boys go climbing. Twice as many girls go sailing as go climbing. Work out how many children go sailing on Sunday.

YOUR SOLUTION

05 Split the ratio

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Ratio Toolkit question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\text{girls} = 120 \times \frac{3}{8} = 45$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1H Q5

Split the ratio

$$\text{girls} = 120 \times \frac{3}{8} = 45$$

Split the ratio

$$\frac{16}{25} \times 75 = 48$$

Finish: 57

FORMULA BANK

Add ratio parts, find one part, then multiply. Keep the order of the ratio fixed.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 05

SAME IDEA Use the same first line from Strategy 05.

QUESTION

5 120 children go on an activity holiday. The ratio of the number of girls to the number of boys is 3 : 5 On Sunday, all the children either go sailing or go climbing. 16 25 of the boys go climbing. Twice as many girls go sailing as go climbing. Work out how many children go sailing on Sunday.

YOUR SOLUTION

06 Total sailing

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Ratio Toolkit question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\text{girls} = 120 \times \frac{3}{8} = 45$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1H Q5

Split the ratio

$$\text{girls} = 120 \times \frac{3}{8} = 45$$

Split the ratio

$$\frac{16}{25} \times 75 = 48$$

Finish: 57

FORMULA BANK

Add ratio parts, find one part, then multiply. Keep the order of the ratio fixed.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 06

SAME IDEA Use the same first line from Strategy 06.

QUESTION

5 120 children go on an activity holiday. The ratio of the number of girls to the number of boys is 3 : 5 On Sunday, all the children either go sailing or go climbing. 16 25 of the boys go climbing. Twice as many girls go sailing as go climbing. Work out how many children go sailing on Sunday.

YOUR SOLUTION

Quick Reference

TRIGGER → FIRST MOVE

WHEN YOU SEE	WRITE THIS FIRST
"work out", "find", a missing value	Find one share
"work out", "show", "hence", a familiar exam phrase	Use total parts
"work out", "show", "hence", a familiar exam phrase	Convert mixed information to parts
"work out", "show", "hence", a familiar exam phrase	Check the order of the ratio
"work out", "show", "hence", a familiar exam phrase	Split the ratio
"work out", "show", "hence", a familiar exam phrase	Total sailing

FINAL BOTTOM LINE Do not try to remember every past-paper wording. Remember the trigger, write the first line, and let the working follow.

ELITE IGCSE MATHEMATICS
EXPERIENCE NOTES

Ratio Problem Solving

Learn the trigger, write the first line, then solve one similar problem while the method is still fresh.

RATIO PROBLEM SOLVING · CH1 · L13



ENG. ESLAM AHMED

The Map

This lesson collapses Ratio Problem Solving into the few exam moves that keep repeating. Read the trigger, copy the first line, then practise the same idea on the next page.

01 Find one share

TRIGGER *"work out", "find", a missing value*

02 Use total parts

TRIGGER *"probability", "random", "given that"*

03 Convert mixed information to parts

TRIGGER *"probability", "random", "given that"*

04 Check the order of the ratio

TRIGGER *"probability", "random", "given that"*

05 Split the ratio

TRIGGER *"probability", "random", "given that"*

06 The ratio is

TRIGGER *"probability", "random", "given that"*

BANK EVIDENCE 7 website questions, 7 checked solutions, 3 local PDFs read.

01 Find one share

TRIGGER *"work out", "find", a missing value*

BECOMES A Ratio Problem Solving question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\text{Brendon : Asha : Julie} = 3 : 2 : 6$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1HR Q1

The ratio is

$$\text{Brendon : Asha : Julie} = 3 : 2 : 6$$

Asha and Julie together have

$$2 + 6 = 8 \text{ parts}$$

Finish: Brendon receives \$13.50.

FORMULA BANK

Turn every sentence into shares or equations before calculating.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 01.

QUESTION

1 Brendon, Asha and Julie share some money in the ratios 3 : 2 : 6 The total amount of money that Asha and Julie receive is \$36 Work out the amount of money that Brendon receives. \$ 1 5 2

YOUR SOLUTION

02 Use total parts

TRIGGER *"probability", "random", "given that"*

BECOMES A Ratio Problem Solving question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\text{Brendon : Asha : Julie} = 3 : 2 : 6$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1HR Q1

The ratio is

$$\text{Brendon : Asha : Julie} = 3 : 2 : 6$$

Asha and Julie together have

$$2 + 6 = 8 \text{ parts}$$

Finish: Brendon receives \$13.50.

FORMULA BANK

Turn every sentence into shares or equations before calculating.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

03 Convert mixed information to parts

TRIGGER *"probability", "random", "given that"*

BECOMES A Ratio Problem Solving question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\text{Danil : Gabriel : Hadley} = 3 : 5 : 9$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2022 P1H Q2

The ratio is

$$\text{Danil : Gabriel : Hadley} = 3 : 5 : 9$$

Split the ratio

$$9 - 5 = 4 \text{ parts}$$

Finish: Danil receives 147 euros.

FORMULA BANK

Turn every sentence into shares or equations before calculating.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 03

SAME IDEA Use the same first line from Strategy 03.

QUESTION

2 Danil, Gabriel and Hadley share some money in the ratios $3 : 5 : 9$ The difference between the amount of money that Gabriel receives and the amount of money that Hadley receives is 196 euros. Work out the amount of money that Danil receives. euros

YOUR SOLUTION

04 Check the order of the ratio

TRIGGER *"probability", "random", "given that"*

BECOMES A Ratio Problem Solving question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\text{Brendon : Asha : Julie} = 3 : 2 : 6$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1HR Q1

The ratio is

$$\text{Brendon : Asha : Julie} = 3 : 2 : 6$$

Asha and Julie together have

$$2 + 6 = 8 \text{ parts}$$

Finish: Brendon receives \$13.50.

FORMULA BANK

Turn every sentence into shares or equations before calculating.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

05 Split the ratio

TRIGGER *"probability", "random", "given that"*

BECOMES A Ratio Problem Solving question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\frac{7}{10} \times 20160 = 14112$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2021 P1H Q1
of the money is changed

$$\frac{7}{10} \times 20160 = 14112$$

Split the ratio

$$\text{dollars} : \text{rupees} : \text{francs} = 9 : 5 : 2$$

Finish: 6174 rand more is changed into dollars than francs.

FORMULA BANK

Turn every sentence into shares or equations before calculating.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 05

SAME IDEA Use the same first line from Strategy 05.

QUESTION

1 Pieter owns a currency conversion shop. Last Monday, Pieter changed a total of 20 160 rand into a number of different currencies. He changed 3 10 of the 20 160 rand into euros. He changed the rest of the rands into dollars, rupees and francs in the ratios 9 : 5 : 2 Pieter changed more rands into dollars than he changed into francs. Work out how many more. rand

YOUR SOLUTION

06 The ratio is

TRIGGER *"probability", "random", "given that"*

BECOMES A Ratio Problem Solving question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\text{Brendon : Asha : Julie} = 3 : 2 : 6$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1HR Q1

The ratio is

$$\text{Brendon : Asha : Julie} = 3 : 2 : 6$$

Asha and Julie together have

$$2 + 6 = 8 \text{ parts}$$

Finish: Brendon receives \$13.50.

FORMULA BANK

Turn every sentence into shares or equations before calculating.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 06

SAME IDEA Use the same first line from Strategy 06.

QUESTION

1 Brendon, Asha and Julie share some money in the ratios 3 : 2 : 6 The total amount of money that Asha and Julie receive is \$36 Work out the amount of money that Brendon receives. \$ 1 5 2

YOUR SOLUTION

Quick Reference

TRIGGER → FIRST MOVE

WHEN YOU SEE	WRITE THIS FIRST
"work out", "find", a missing value	Find one share
"probability", "random", "given that"	Use total parts
"probability", "random", "given that"	Convert mixed information to parts
"probability", "random", "given that"	Check the order of the ratio
"probability", "random", "given that"	Split the ratio
"probability", "random", "given that"	The ratio is

FINAL BOTTOM LINE Do not try to remember every past-paper wording. Remember the

trigger, write the first line, and let the working follow.

ELITE IGCSE MATHEMATICS
EXPERIENCE NOTES

Exchange Rates & Best Buys

Learn the trigger, write the first line, then solve one similar problem while the method is still fresh.

EXCHANGE RATES & BEST BUYS · CH1 · L14



ENG. ESLAM AHMED

The Map

This lesson collapses Exchange Rates & Best Buys into the few exam moves that keep repeating. Read the trigger, copy the first line, then practise the same idea on the next page.

01 Write the exchange direction

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

02 Convert to one currency

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

03 Compare unit prices

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

04 Round money sensibly

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

05 Convert standard form

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

06 litres

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BANK EVIDENCE 1 website questions, 1 checked solutions, 3 local PDFs read.

01 Write the exchange direction

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Exchange Rates & Best Buys question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$4.2 \times 10^5 = 420000$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P2H Q12

Convert standard form

$$4.2 \times 10^5 = 420000$$

litres

$$\text{Dane cost per litre} = \frac{2500000}{420000} = 5.952 \dots$$

Finish: Arctic Oil

FORMULA BANK

Write the direction of exchange first. Best buy compares same quantity or same money.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 01

SAME IDEA Use the same first line from Strategy 01.

QUESTION

12 Astrid wants to buy some oil. She can buy the oil from either Dane Oil or Arctic Oil. Here is information about the price that each company will charge Astrid. Dane Oil Arctic Oil (4.2 times 10⁵) litres for 2 500 000 Krone (8.6 times 10⁵) litres for 770 000 Dollars Astrid wants to get the better value for money for the oil. 1 Dollar = 6.57 Krone From which company should she buy her oil, Dane Oil or Arctic Oil? You must show your working.

YOUR SOLUTION

02 Convert to one currency

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Exchange Rates & Best Buys question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$4.2 \times 10^5 = 420000$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P2H Q12

Convert standard form

$$4.2 \times 10^5 = 420000$$

litres

$$\text{Dane cost per litre} = \frac{2500000}{420000} = 5.952 \dots$$

Finish: Arctic Oil

FORMULA BANK

Write the direction of exchange first. Best buy compares same quantity or same money.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 02

SAME IDEA Use the same first line from Strategy 02.

QUESTION

12 Astrid wants to buy some oil. She can buy the oil from either Dane Oil or Arctic Oil. Here is information about the price that each company will charge Astrid.

	Dane Oil	Arctic Oil
(4.2 times 10 ⁵) litres for	2 500 000 Krone	(8.6 times 10 ⁵) litres for
770 000 Dollars		

Astrid wants to get the better value for money for the oil. 1 Dollar = 6.57 Krone

From which company should she buy her oil, Dane Oil or Arctic Oil? You must show your working.

YOUR SOLUTION

03 Compare unit prices

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Exchange Rates & Best Buys question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$4.2 \times 10^5 = 420000$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P2H Q12

Convert standard form

$$4.2 \times 10^5 = 420000$$

litres

$$\text{Dane cost per litre} = \frac{2500000}{420000} = 5.952 \dots$$

Finish: Arctic Oil

FORMULA BANK

Write the direction of exchange first. Best buy compares same quantity or same money.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 03

SAME IDEA Use the same first line from Strategy 03.

QUESTION

12 Astrid wants to buy some oil. She can buy the oil from either Dane Oil or Arctic Oil. Here is information about the price that each company will charge Astrid. Dane Oil Arctic Oil (4.2 times 10⁵) litres for 2 500 000 Krone (8.6 times 10⁵) litres for 770 000 Dollars Astrid wants to get the better value for money for the oil. 1 Dollar = 6.57 Krone From which company should she buy her oil, Dane Oil or Arctic Oil? You must show your working.

YOUR SOLUTION

04 Round money sensibly

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Exchange Rates & Best Buys question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$4.2 \times 10^5 = 420000$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P2H Q12

Convert standard form

$$4.2 \times 10^5 = 420000$$

litres

$$\text{Dane cost per litre} = \frac{2500000}{420000} = 5.952 \dots$$

Finish: Arctic Oil

FORMULA BANK

Write the direction of exchange first. Best buy compares same quantity or same money.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 04

SAME IDEA Use the same first line from Strategy 04.

QUESTION

12 Astrid wants to buy some oil. She can buy the oil from either Dane Oil or Arctic Oil. Here is information about the price that each company will charge Astrid. Dane Oil Arctic Oil (4.2 times 10⁵) litres for 2 500 000 Krone (8.6 times 10⁵) litres for 770 000 Dollars Astrid wants to get the better value for money for the oil. 1 Dollar = 6.57 Krone From which company should she buy her oil, Dane Oil or Arctic Oil? You must show your working.

YOUR SOLUTION

05 Convert standard form

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Exchange Rates & Best Buys question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$4.2 \times 10^5 = 420000$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P2H Q12

Convert standard form

$$4.2 \times 10^5 = 420000$$

litres

$$\text{Dane cost per litre} = \frac{2500000}{420000} = 5.952 \dots$$

Finish: Arctic Oil

FORMULA BANK

Write the direction of exchange first. Best buy compares same quantity or same money.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 05

SAME IDEA Use the same first line from Strategy 05.

QUESTION

12 Astrid wants to buy some oil. She can buy the oil from either Dane Oil or Arctic Oil. Here is information about the price that each company will charge Astrid. Dane Oil Arctic Oil (4.2 times 10⁵) litres for 2 500 000 Krone (8.6 times 10⁵) litres for 770 000 Dollars Astrid wants to get the better value for money for the oil. 1 Dollar = 6.57 Krone From which company should she buy her oil, Dane Oil or Arctic Oil? You must show your working.

YOUR SOLUTION

06 litres

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Exchange Rates & Best Buys question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$4.2 \times 10^5 = 420000$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P2H Q12

Convert standard form

$$4.2 \times 10^5 = 420000$$

litres

$$\text{Dane cost per litre} = \frac{2500000}{420000} = 5.952 \dots$$

Finish: Arctic Oil

FORMULA BANK

Write the direction of exchange first. Best buy compares same quantity or same money.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 06

SAME IDEA Use the same first line from Strategy 06.

QUESTION

12 Astrid wants to buy some oil. She can buy the oil from either Dane Oil or Arctic Oil. Here is information about the price that each company will charge Astrid. Dane Oil Arctic Oil (4.2 times 10⁵) litres for 2 500 000 Krone (8.6 times 10⁵) litres for 770 000 Dollars Astrid wants to get the better value for money for the oil. 1 Dollar = 6.57 Krone From which company should she buy her oil, Dane Oil or Arctic Oil? You must show your working.

YOUR SOLUTION

Quick Reference

TRIGGER → FIRST MOVE

WHEN YOU SEE	WRITE THIS FIRST
"work out", "show", "hence", a familiar exam phrase	Write the exchange direction
"work out", "show", "hence", a familiar exam phrase	Convert to one currency
"work out", "show", "hence", a familiar exam phrase	Compare unit prices
"work out", "show", "hence", a familiar exam phrase	Round money sensibly
"work out", "show", "hence", a familiar exam phrase	Convert standard form
"work out", "show", "hence", a familiar exam phrase	litres

FINAL BOTTOM LINE Do not try to remember every past-paper wording. Remember the trigger, write the first line, and let the working follow.

ELITE IGCSE MATHEMATICS
EXPERIENCE NOTES

Direct & Inverse Proportion

Learn the trigger, write the first line, then solve one similar problem while the method is still fresh.

DIRECT & INVERSE PROPORTION · CH1 · L15

ENG. ESLAM AHMED



The Map

This lesson collapses Direct & Inverse Proportion into the few exam moves that keep repeating. Read the trigger, copy the first line, then practise the same idea on the next page.

01 Choose direct or inverse

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

02 Find the constant k

TRIGGER *"work out", "find", a missing value*

03 Substitute the new value

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

04 Check the direction of change

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

05 Find inverse function

TRIGGER *"work out", "find", a missing value*

06 Use when

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BANK EVIDENCE 16 website questions, 16 checked solutions, 3 local PDFs read.

01 Choose direct or inverse

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Direct & Inverse Proportion question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$A = \frac{k}{C^2}$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2022 P1H Q15

Find inverse function

$$A = \frac{k}{C^2}$$

Use when

$$40 = \frac{k}{1.5^2}$$

Finish: $C = 0.3$.

FORMULA BANK

Direct: $y = kx$. Inverse: $y = \frac{k}{x}$. Find k first.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

02 Find the constant k

TRIGGER *"work out", "find", a missing value*

BECOMES A Direct & Inverse Proportion question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$y \propto \frac{1}{\sqrt{x}}$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2022 P2HR Q20

Find inverse function

$$y \propto \frac{1}{\sqrt{x}}$$

Find inverse function

This is a reasoning step: write one clear sentence, then continue the calculation.

Finish: $T = \frac{100}{9}$.

FORMULA BANK

Direct: $y = kx$. Inverse: $y = \frac{k}{x}$. Find k first.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 02

SAME IDEA Use the same first line from Strategy 02.

QUESTION

20 y is inversely proportional to x x is directly proportional to T 3 Given that y = 8 when T = 25 find the exact value of T when y = 27 T =

YOUR SOLUTION

03 Substitute the new value

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Direct & Inverse Proportion question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$A = \frac{k}{C^2}$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2022 P1H Q15

Find inverse function

$$A = \frac{k}{C^2}$$

Use when

$$40 = \frac{k}{1.5^2}$$

Finish: $C = 0.3$.

FORMULA BANK

Direct: $y = kx$. Inverse: $y = \frac{k}{x}$. Find k first.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 03

SAME IDEA Use the same first line from Strategy 03.

QUESTION

15 A is inversely proportional to C
A = 40 when C = 1.5
Calculate the value of C when A = 1000
C =

YOUR SOLUTION

04 Check the direction of change

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Direct & Inverse Proportion question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$M = kh^3$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jun 2022 P2H Q17

Find inverse function

$$M = kh^3$$

Use when

$$4 = k(0.5)^3$$

Finish: $h = 2.5$.

FORMULA BANK

Direct: $y = kx$. Inverse: $y = \frac{k}{x}$. Find k first.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

05 Find inverse function

TRIGGER *"work out", "find", a missing value*

BECOMES A Direct & Inverse Proportion question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$A = \frac{k}{C^2}$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2022 P1H Q15

Find inverse function

$$A = \frac{k}{C^2}$$

Use when

$$40 = \frac{k}{1.5^2}$$

Finish: $C = 0.3$.

FORMULA BANK

Direct: $y = kx$. Inverse: $y = \frac{k}{x}$. Find k first.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

06 Use when

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Direct & Inverse Proportion question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$A = \frac{k}{C^2}$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2022 P1H Q15

Find inverse function

$$A = \frac{k}{C^2}$$

Use when

$$40 = \frac{k}{1.5^2}$$

Finish: $C = 0.3$.

FORMULA BANK

Direct: $y = kx$. Inverse: $y = \frac{k}{x}$. Find k first.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Quick Reference

TRIGGER → FIRST MOVE

WHEN YOU SEE	WRITE THIS FIRST
"work out", "show", "hence", a familiar exam phrase	Choose direct or inverse
"work out", "find", a missing value	Find the constant k
"work out", "show", "hence", a familiar exam phrase	Substitute the new value
"work out", "show", "hence", a familiar exam phrase	Check the direction of change
"work out", "find", a missing value	Find inverse function
"work out", "show", "hence", a familiar exam phrase	Use when

FINAL BOTTOM LINE Do not try to remember every past-paper wording. Remember the trigger, write the first line, and let the working follow.